Linear Algebra | Quiz for §4.4 &4.5 | October 23, 2026

Name: _____ R. Hammack Score: _____

Directions: Show and explain all work. Put phones away. No calculators, etc.

1. Suppose $\mathbf{u}, \mathbf{v}, \mathbf{w}$ are three vectors in a vector space V. Without knowing any further information, is it possible to say whether or not the set $\{\mathbf{v}-\mathbf{u},\ \mathbf{w}-\mathbf{v},\ \mathbf{u}-\mathbf{w}\}$ is linearly independent or dependent? Explain.

2. Find a basis for the subspace $W = \{(2s-t, s, t)\ :\ s, t \in \mathbb{R}\} \subseteq \mathbb{R}^3$.

3. Is the set $S = \left\{ \begin{bmatrix} 3 & 5 \\ 2 & -2 \end{bmatrix}, \begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix} \right\} \subseteq M_{2,2}$ linearly independent or dependent?